



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO Sampoorina Agencies Unit II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

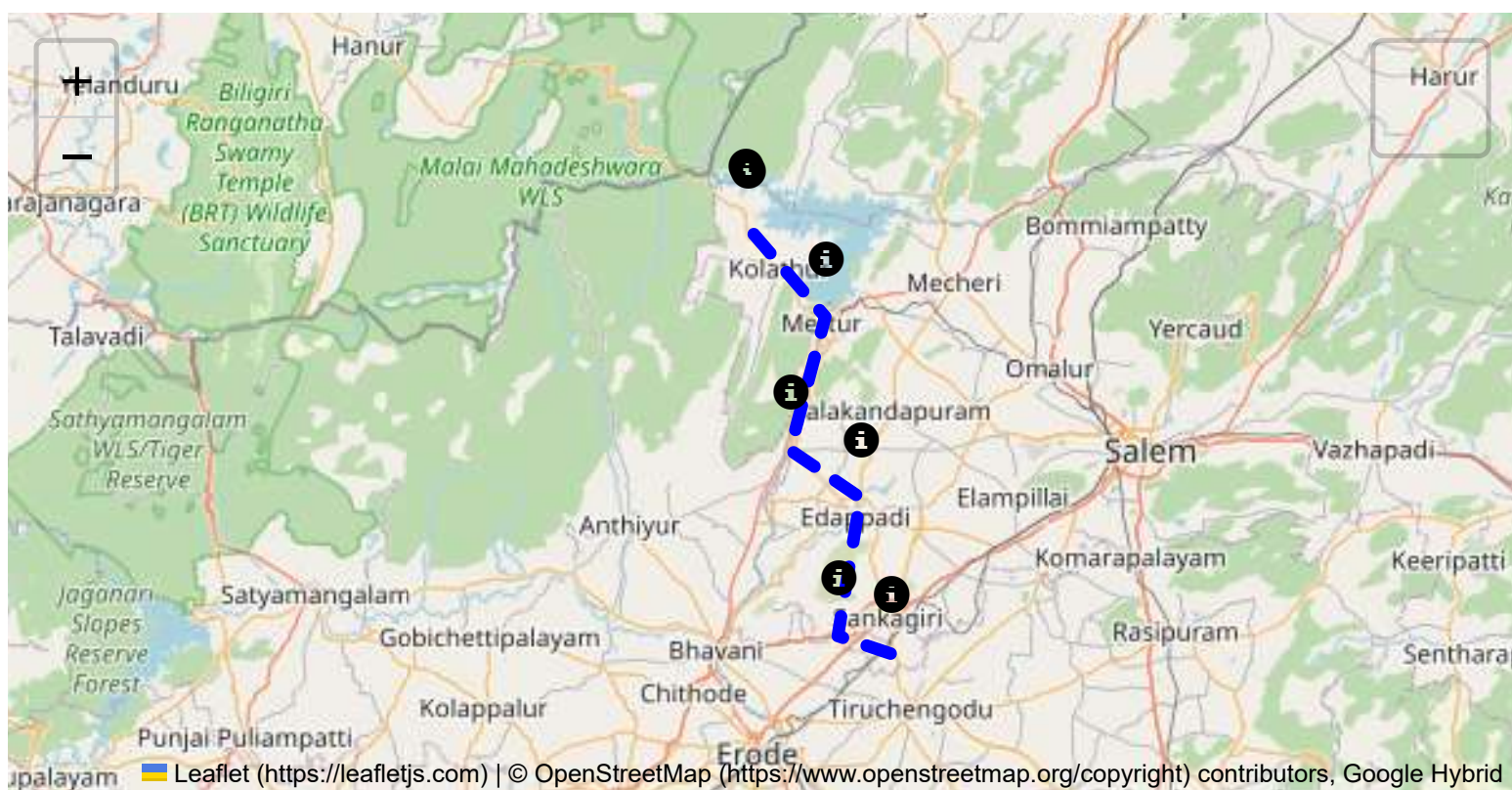
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

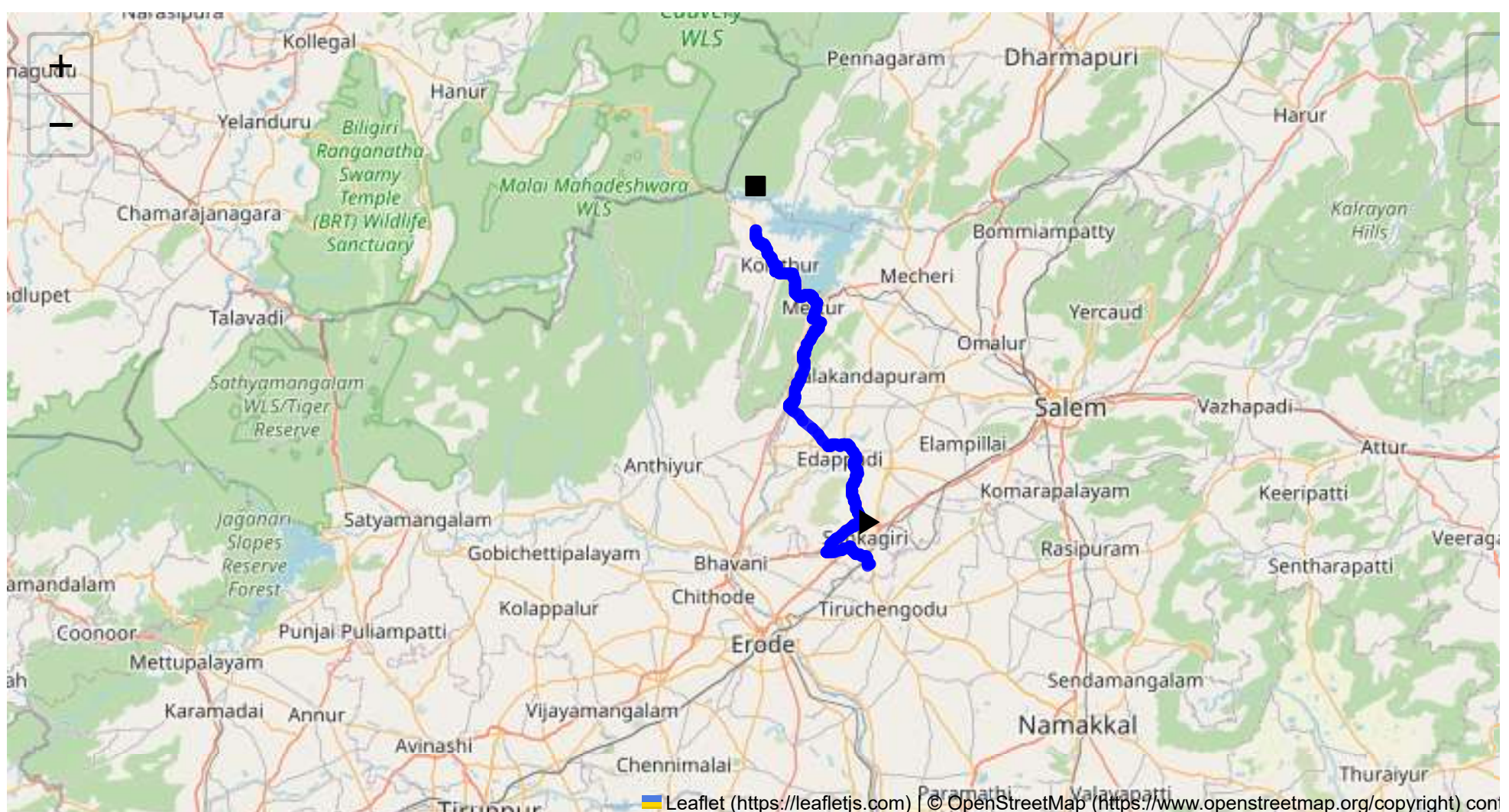
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 77.90 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (11.895912, 77.715714)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to Major District Road 268 Palavadi spans approximately 77.90 kilometers, passing through several towns such as Puthur, Padaiveedu, Idappadi, and Mettur. It primarily involves traveling along main district roads which are connected by several minor roads and bypasses. The route typically takes around 1.90 hours for heavy vehicles carrying hazardous materials.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with high temperatures and significant rainfall during monsoon seasons (June to September). The route may be affected by heavy rains leading to potential flooding and poor road visibility. Travelers should be cautious during these times to avoid hydroplaning and slippery surfaces.

3. Analysis of Traffic Patterns

Traffic patterns can vary, with peak congestion typically occurring during morning and evening hours (8:00-10:00 AM and 5:00-8:00 PM). Particularly busy areas may include towns like Idappadi and Mettur, where local traffic can be dense.

4. Assessment of Road Quality and Infrastructure

The route primarily consists of district roads, which may vary in quality. Some sections might be well-paved, but others, especially in rural and less developed areas, could have potholes, uneven surfaces, or lack adequate lighting.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, one option is to divert traffic towards National Highway 544, which runs parallel to the route and offers higher quality roads and better facilities. Alternatively, NH79 provides access to major towns and better emergency response infrastructure.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is subject to strict regulations in Tamil Nadu, including requirements for safety equipment, proper labeling, and permissions. Drivers must ensure compliance with local laws, such as the Central Motor Vehicles Rules, which mandate emergency information and firefighting apparatus aboard the vehicle.

7. Overview of Historical Incidents

While specific historical incidents are sparse, typical risks involve accidents due to road conditions or driver fatigue. It is essential to maintain high safety standards and periodic checks to avoid such occurrences.

8. Environmental Considerations and Sensitive Areas

The route may pass through environmentally sensitive areas, particularly near water bodies close to Mettur. Transporters must ensure no spillage occurs to protect local wildlife and habitats.

9. Analysis of Communication Coverage

Most of the route has good communication coverage with occasional dead zones, particularly in rural and less populated areas between towns. Provisions for satellite phones or offline navigation tools are recommended for continuous communication.

10. Estimated Emergency Response Times

Emergency response time can vary:

- In urbanized segments (e.g., Mettur): 20-30 minutes.
- Rural regions: 40-60 minutes. Routes closer to NH roads tend to have better emergency services available.

12. Overall Summary of Risk Assessment

Overall, while the route is serviceable for the transport of hazardous materials, drivers should remain vigilant due to variable traffic patterns and road conditions. Weather conditions, particularly during the monsoon, pose significant hazards. Compliance with local regulations and maintaining constant communication will mitigate risks. It's essential for drivers and companies to conduct regular safety training and equipment checks to ensure safety and efficiency.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60019, 77.81968	15 KM/Hr
2	Roundabout	High	11.77576, 77.79789	15 KM/Hr
3	Turn	High	11.43961, 77.87341	15 KM/Hr
4	Turn	High	11.44024, 77.87528	15 KM/Hr
5	Turn	High	11.44898, 77.87410	15 KM/Hr
6	Turn	High	11.45352, 77.85698	15 KM/Hr
7	Turn	Medium	11.45847, 77.85140	30 KM/Hr
8	Turn	Medium	11.45898, 77.85135	30 KM/Hr
9	Turn	High	11.46303, 77.84945	15 KM/Hr
10	Turn	Medium	11.45543, 77.81740	30 KM/Hr
11	Turn	High	11.45631, 77.81668	15 KM/Hr
12	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
13	Turn	Medium	11.49433, 77.86696	30 KM/Hr
14	Turn	Medium	11.50874, 77.86045	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	Medium	11.52797, 77.85580	30 KM/Hr
16	Turn	Medium	11.54763, 77.85279	30 KM/Hr
17	Turn	Medium	11.55490, 77.85647	30 KM/Hr
18	Turn	Medium	11.56416, 77.85968	30 KM/Hr
19	Turn	High	11.57903, 77.85432	15 KM/Hr
20	Turn	Medium	11.57917, 77.85578	30 KM/Hr
21	Turn	Medium	11.58025, 77.85746	30 KM/Hr
22	Turn	Medium	11.58672, 77.85835	30 KM/Hr
23	Turn	Medium	11.60232, 77.83510	30 KM/Hr
24	Turn	High	11.65395, 77.76448	15 KM/Hr
25	Turn	Medium	11.65683, 77.76463	30 KM/Hr
26	Turn	High	11.65758, 77.76508	15 KM/Hr
27	Turn	Medium	11.65795, 77.76624	30 KM/Hr
28	Turn	High	11.66102, 77.76755	15 KM/Hr
29	Turn	High	11.66089, 77.76847	15 KM/Hr
30	Turn	Medium	11.66121, 77.76883	30 KM/Hr
31	Turn	High	11.66121, 77.76991	15 KM/Hr
32	Turn	High	11.66385, 77.77082	15 KM/Hr
33	Turn	High	11.66397, 77.77135	15 KM/Hr
34	Turn	High	11.66855, 77.77232	15 KM/Hr
35	Turn	High	11.66872, 77.77192	15 KM/Hr
36	Turn	High	11.66911, 77.77183	15 KM/Hr
37	Turn	High	11.66943, 77.77209	15 KM/Hr
38	Turn	Medium	11.72624, 77.78413	30 KM/Hr
39	Turn	High	11.72742, 77.78410	15 KM/Hr
40	Turn	Medium	11.72745, 77.78513	30 KM/Hr
41	Turn	High	11.72780, 77.78579	15 KM/Hr
42	Turn	High	11.73934, 77.78806	15 KM/Hr
43	Turn	High	11.75593, 77.79870	15 KM/Hr
44	Turn	Medium	11.75670, 77.79842	30 KM/Hr
45	Turn	High	11.75721, 77.79857	15 KM/Hr
46	Turn	Medium	11.75750, 77.79991	30 KM/Hr
47	Turn	Medium	11.76228, 77.80499	30 KM/Hr
48	Turn	High	11.76306, 77.80533	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
49	Turn	High	11.76398, 77.80428	15 KM/Hr
50	Turn	Medium	11.77011, 77.80793	30 KM/Hr
51	Turn	High	11.77053, 77.80810	15 KM/Hr
52	Turn	High	11.79763, 77.80269	15 KM/Hr
53	Turn	Medium	11.79797, 77.80234	30 KM/Hr
54	Turn	Medium	11.79809, 77.80211	30 KM/Hr
55	Turn	High	11.80264, 77.79727	15 KM/Hr
56	Turn	High	11.80340, 77.79754	15 KM/Hr
57	Turn	Medium	11.80348, 77.79746	30 KM/Hr
58	Turn	High	11.80672, 77.79707	15 KM/Hr
59	Turn	Medium	11.80790, 77.79264	30 KM/Hr
60	Turn	Medium	11.80914, 77.79055	30 KM/Hr
61	Turn	Medium	11.80835, 77.78581	30 KM/Hr
62	Turn	Medium	11.80506, 77.78073	30 KM/Hr
63	Turn	Medium	11.81022, 77.77339	30 KM/Hr
64	Turn	Medium	11.81538, 77.77208	30 KM/Hr
65	Turn	Medium	11.81943, 77.77231	30 KM/Hr
66	Turn	Medium	11.82111, 77.77150	30 KM/Hr
67	Turn	Medium	11.82582, 77.77298	30 KM/Hr
68	Turn	Medium	11.83627, 77.76936	30 KM/Hr
69	Turn	Medium	11.83800, 77.76744	30 KM/Hr
70	Turn	Medium	11.83786, 77.75154	30 KM/Hr
71	Turn	Medium	11.83866, 77.75068	30 KM/Hr
72	Turn	Medium	11.83980, 77.74731	30 KM/Hr
73	Turn	Medium	11.83979, 77.74727	30 KM/Hr
74	Turn	Medium	11.84066, 77.74477	30 KM/Hr
75	Turn	Medium	11.84117, 77.74460	30 KM/Hr
76	Turn	High	11.84733, 77.74614	15 KM/Hr
77	Turn	Medium	11.85202, 77.74197	30 KM/Hr
78	Turn	Medium	11.87460, 77.73003	30 KM/Hr
79	Turn	Medium	11.87521, 77.72891	30 KM/Hr
80	Turn	Medium	11.87573, 77.72853	30 KM/Hr
81	Turn	High	11.87734, 77.72847	15 KM/Hr
82	Turn	Medium	11.87995, 77.72327	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
83	Turn	Medium	11.88080, 77.72287	30 KM/Hr
84	Turn	Medium	11.88154, 77.72021	30 KM/Hr
85	Turn	Medium	11.88862, 77.71715	30 KM/Hr
86	Turn	Medium	11.89053, 77.71742	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	S.K.A Hospital	11.7816111, 77.797688	30 km/h	Medium
3	hospital	Sathya Hospital	11.7864589, 77.8004201	30 km/h	Medium
4	clinic	Saraswathi Poly Clinic	11.7905491, 77.8007345	30 km/h	Medium
6	clinic	Duraisamy Eye Clinic	11.7942636, 77.7995843	30 km/h	Medium
7	clinic	Shiva Siddha Clinic	11.794237, 77.7995531	30 km/h	Medium
8	hospital	Govt Hospital Mettur	11.7959119, 77.7987959	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Govt. Arts College, Mettur	11.7756806, 77.7958483	30 km/h	Medium
5	school	Govt. Hr. Sec. School	11.792309, 77.8006337	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60019, 77.81968



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77576, 77.79789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44024, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49390, 77.86748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49433, 77.86696



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50874, 77.86045



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52797, 77.85580



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54763, 77.85279



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55490, 77.85647



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56416, 77.85968



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57903, 77.85432



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57917, 77.85578



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58025, 77.85746



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58672, 77.85835



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60232, 77.83510



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65395, 77.76448



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65683, 77.76463



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65758, 77.76508



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65795, 77.76624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66102, 77.76755



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66089, 77.76847



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66121, 77.76883



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66121, 77.76991



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66385, 77.77082



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66397, 77.77135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66855, 77.77232



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66872, 77.77192



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66911, 77.77183



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66943, 77.77209



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72624, 77.78413



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72742, 77.78410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72745, 77.78513



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.72780, 77.78579



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73934, 77.78806



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75593, 77.79870



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75670, 77.79842



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75721, 77.79857



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75750, 77.79991



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76228, 77.80499



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76306, 77.80533



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76398, 77.80428



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77011, 77.80793



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.77053, 77.80810



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.79763, 77.80269



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.79797, 77.80234



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.79809, 77.80211



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.80264, 77.79727



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.80340, 77.79754



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80348, 77.79746



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.80672, 77.79707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80790, 77.79264



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80914, 77.79055



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80835, 77.78581



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80506, 77.78073



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81022, 77.77339



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81538, 77.77208



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81943, 77.77231



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.82111, 77.77150



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

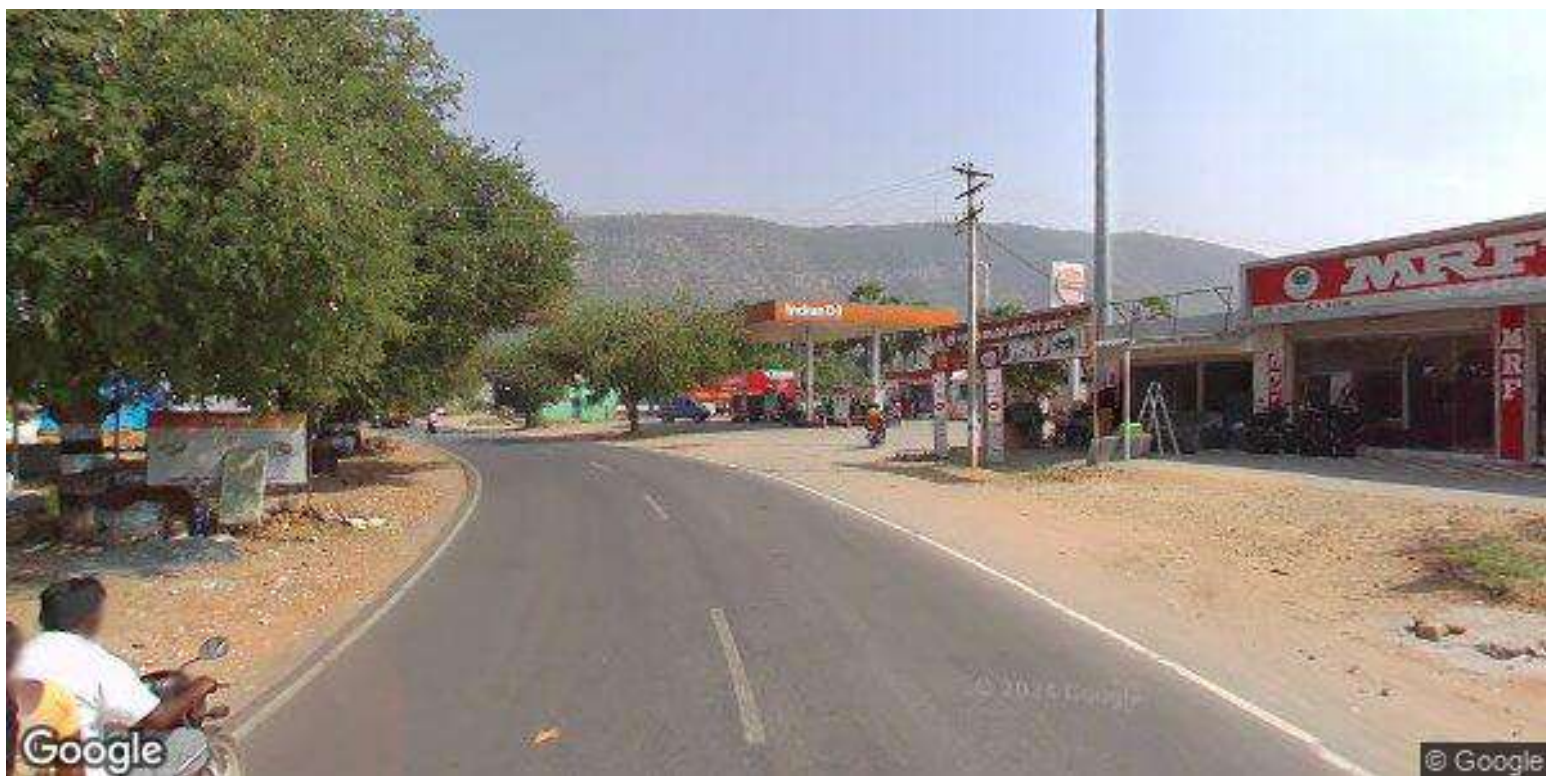
Coordinates: 11.82582, 77.77298



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83627, 77.76936



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83800, 77.76744



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83786, 77.75154



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83866, 77.75068



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83980, 77.74731



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83979, 77.74727



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84066, 77.74477



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84117, 77.74460



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.84733, 77.74614



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.85202, 77.74197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87460, 77.73003



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87521, 77.72891



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87573, 77.72853



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.87734, 77.72847



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87995, 77.72327



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88080, 77.72287



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88154, 77.72021



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88862, 77.71715



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.89053, 77.71742
